2 A boulder (supported by cable QR) hangs from a uniform metal rod PQ, freely pivoted at P as shown in Fig. 2.1. The rod is 6.0 m long and has a mass of 15 kg. The mass of the boulder is 17 kg. A horizontal cable XY supports the rod and ensures that the rod is in equilibrium.

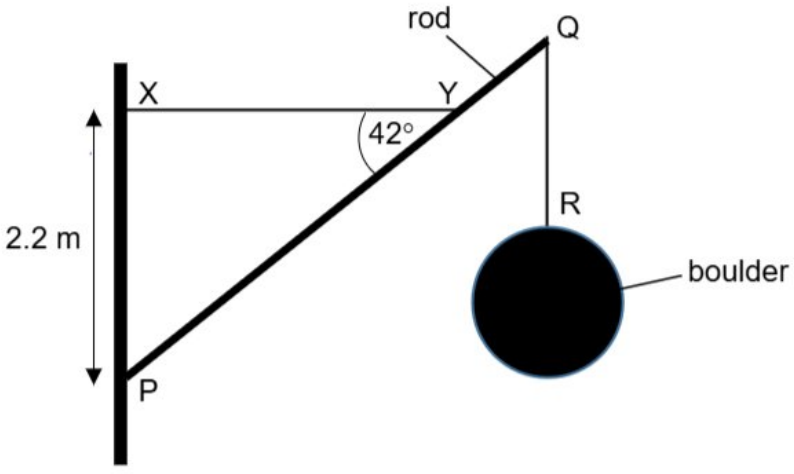


Fig. 2.1

(a) State the conditions required for the rod to be in equilibrium.

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.....

..... [2]

(b) On Fig. 2.1, draw all the forces acting on rod PQ and label them clearly. [2]

(c) Determine the tension in the horizontal cable XY. Assume the mass of cable XY to be negligible.

tension in cable XY = ... N [2]

(d) The boulder in Fig. 2.1 is replaced with a crate that has a larger mass.

Without doing any calculations, state and explain how the tension in cable XY would change. Assume cable XY remains horizontal.

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..... [2]

[Total:8]